Voice Control Stacking

Note: Due to unavoidable servo mechanical errors and the influence of servo center position, if the grabbing and placing positions are not ideal, you can try recalibrating the robotic arm or modify the grabbing and placing positions according to your needs.

1. Function Description

Through voice commands, specify the color of the block to grab. The robotic arm will grab the corresponding color block and stack them layer by layer.

2. Startup and Operation

2.1 Startup

Open the terminal and enter the following command to start:

```
python3 ~/dofbot_voice/scripts/voice_control_stack.py
```

2.2 Operation Steps

After the program runs successfully, place the blocks in positions: yellow area for yellow blocks, red area for red blocks, green area for green blocks, blue area for blue blocks, as shown in the figure below. Unlike previous stacking cases, this case allows you to specify the order of stacking.

2.2.1 Stacking

Say "Hi, yahboom" to the voice module. The voice module replies "I'm here" indicating successful wake-up. Then say "Stack the yellow block"/"Stack the red block"/"Stack the green block"/"Stack the blue block", the voice module will broadcast "ok". The robotic arm will go to the corresponding color block area, grab the block, and place it in the set area. The voice module will broadcast "Placement complete". Each time you use a voice command to grab the target color block and place it, and finally complete the stacking.

2.2.2 Pushing Down

Say "Hi, yahboom" to the voice module. The voice module replies "I'm here" indicating successful wake-up. Then say "Push down", the voice module will broadcast "ok". Then the robotic arm will execute the action to push down the stacked blocks.

3. Core Code Analysis

Jetson-Nano users need to enter the docker container to view.

`Source code path: ~/dofbot_voice/scripts/voice_control_stack.py

```
    while True:
        # Get voice recognition result
        res = mySpeech.speech_read()
        time.sleep(0.01)
        # Grab yellow block
        if res == 77:
            if yellow_grabbed == 0:
                mySpeech.void_write(77)
                Arm.Arm_Buzzer_On(1)
                time.sleep(.1)
                start_move_arm(1)
                time.sleep(1)
                # global yellow_grabbed
                yellow_grabbed = 1
        # Grab red block
        elif res == 78:
            if red_grabbed == 0:
                mySpeech.void_write(77)
                Arm.Arm_Buzzer_On(1)
                time.sleep(.1)
                start_move_arm(2)
                # global red_grabbed
                red_grabbed = 1
        # Grab green block
        elif res == 79:
            if green_grabbed == 0:
                mySpeech.void_write(77)
                Arm.Arm_Buzzer_On(1)
                time.sleep(.1)
                start_move_arm(3)
                # global green_grabbed
                green_grabbed = 1
        # Grab blue block
        elif res == 80:
            if blue_grabbed == 0:
                mySpeech.void_write(77)
```

```
                Arm.Arm_Buzzer_On(1)
                time.sleep(.1)
                start_move_arm(4)
                # global blue_grabbed
                blue_grabbed = 1
        elif res == 82: # Push down
            mySpeech.void_write(82)
            Arm.Arm_Buzzer_On(1)
            time.sleep(.1)
            start_move_arm(5)
            yellow_grabbed = 0
            red_grabbed = 0
            green_grabbed = 0
            blue_grabbed = 0


# Number function definition
def number_action(index):
    if index == 1:
        # Grab yellow block
        arm_move(p_top, 1000)
        arm_move(p_Yellow, 1000)
        arm_clamp_block(1)
        Arm.Arm_serial_servo_write(2, 90, 1000)
        time.sleep(1)
        arm_move(p_top, 1000)
    elif index == 2:
        # Grab red block
        arm_move(p_top, 1000)
        arm_move(p_Red, 1000)
        arm_clamp_block(1)
        Arm.Arm_serial_servo_write(2, 90, 1000)
        time.sleep(1)
        arm_move(p_top, 1000)
    elif index == 3:
        # Grab green block
        arm_move(p_top, 1000)
        arm_move(p_Green, 1000)
        arm_clamp_block(1)
        Arm.Arm_serial_servo_write(2, 90, 1000)
        time.sleep(1)
        arm_move(p_top, 1000)
    elif index == 4:
        # Grab blue block
        arm_move(p_top, 1000)
        arm_move(p_Blue, 1000)
        arm_clamp_block(1)
        Arm.Arm_serial_servo_write(2, 90, 1000)
        time.sleep(1)
        arm_move(p_top, 1000)

# Place stacked blocks, layer indicates which layer to place on
def put_down_block(layer):
    if layer == 1:
        arm_move(p_layer_1, 1000)
        arm_clamp_block(0)
        arm_move_6(look_at, 1000)
    elif layer == 2:
```

```python
        arm_move(p_layer_2, 1000)
        arm_clamp_block(0)
        arm_move_6(look_at, 1000)
    elif layer == 3:
        arm_move(p_layer_3, 1000)
        arm_clamp_block(0)
        arm_move_6(look_at, 1000)
    elif layer == 4:
        arm_move(p_layer_4, 1000)
        time.sleep(.1)
        arm_clamp_block(0)
        arm_move_6(look_at, 1000)
    # Voice broadcast, placement complete
    mySpeech.void_write(81)
```